

HARIPRASAD BATHINI SANKARAN

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AWS DATA ENGINEER

SUMMARY

Data Engineer with 4+ years of experience designing large-scale pipelines, ETL/ELT workflows, and secure data solutions using **Python**, **PySpark**, and **SQL** on **AWS**. Proficient in batch and real-time streaming using **AWS Glue**, **Lambda**, **Kinesis**, and **EMR**, with expertise in **S3**-based Data Lakes, **Redshift**, and **Terraform CI/CD**. Experienced in **data modeling** and pipeline development across **security** and **healthcare** domains with advanced analytics.

SKILLS

Data Engineering: Python, SQL, PySpark, ETL/ELT, Data Modeling, Data Pipelines, Data Warehousing, Medallion Architecture, Data Lake
Big Data & Processing: AWS Glue, EMR, Apache Spark, Delta Lake, Kinesis Streaming, Batch Processing, Redshift, PostgreSQL
Business Intelligence: AWS QuickSight, Operational Analytics, Data Visualization, Reporting, Database
Cloud & DevOps: AWS (S3, Lambda, Glue, EMR, Kinesis, Redshift, Athena, IAM, CloudWatch), Terraform, CI/CD, Git, GitHub
Tools & Practices: VS Code, Jupyter, Unix/Linux, Agile, Documentation, Version Control, AI Agents, HL7, FHIR, Data Structures
Generative AI & LLMs: Prompt Engineering, Cursor, GitHub Copilot, Gemini CLI, AI Assisted Development, Statistics, Mathematics

WORK EXPERIENCE

Data Engineer | Comcast Corporation | Chennai, India **Aug 2023 – Aug 2025**

- Architected and maintained **AWS Glue** and **EMR** pipelines processing **1TB** of daily **cybersecurity** logs from **S3** for threat detection.
- Automated **AWS Glue** orchestration workflows for 10+ ETL pipelines with 5 concurrent executions, ensuring zero dependency failures.
- Developed **modular, parameterized PySpark** scripts to standardize data cleansing and accelerate new telemetry source onboarding.
- Optimized storage and compute using **S3 partitioning**, Glue job bookmarks, and columnar formats, reducing **AWS** compute costs by **20%**.
- Managed **AWS IAM** and **Lake Formation** policies via **Terraform**, enforcing role-based access controls (RBAC) across teams.
- Accelerated ETL pipeline development using GenAI tools (**GitHub Copilot**), reducing **AWS** deployment time from 10 days to 7 days.
- Mentoring junior team members on pipeline architecture and coding standards with technical documentation to speed up their involvement.
- Monitored pipeline health and **data quality** metrics using **CloudWatch**, providing timely resolutions to ensure SLA compliance.

Data Engineer | Infosys Ltd. | Chennai, India **Jun 2021 – Aug 2023**

- Developed ETL pipelines integrating raw operational datasets into analytics-ready layers in **S3** and **Redshift** using **AWS Glue**.
- Migrated complex T-SQL stored procedures to distributed **PySpark** workflows on **EMR**, reducing batch processing time by **30 mins/run**.
- Designed metadata-driven ingestion frameworks using **AWS Glue Data Catalog** to support scalable onboarding of new data sources.
- Built **Python** REST API ingestion frameworks with **Lambda**, retry logic, and pagination, reducing integration time from weeks to days.
- Constructed multi-dimensional **data models** (fact and dimension tables) in **Redshift** to support BI dashboards and reporting.
- Established automated **data quality** checks using **Python** and **AWS Glue** to detect schema drift and prevent corrupted downstream.
- Partnered with **BI analysts** to design and deliver optimized **Redshift** tables, powering operational dashboards and reducing query latency.

PROJECTS

Security Log Analytics and Threat Detection Platform (AWS)

- Engineered scalable data pipelines using **AWS Glue** and **EMR** to ingest and process security data from simulated **S3** sources.
- Developed transformation workflows with **PySpark** and stored curated datasets in **S3** using Parquet and Delta formats.
- Architected **medallion architecture** to enable operational analytics and support historical reporting workflows with automation.
- Implemented **data validation** checks and **CloudWatch** monitoring logic to assess pipeline reliability and ensure **data quality** in production.

Hybrid Healthcare Data Engineering Pipeline (FHIR → Medallion → RBAC) (Ongoing)

- Engineered end-to-end **medallion architecture (Bronze → Silver → Gold)** ingesting **synthetic FHIR JSON bundles** using **PySpark**.
- Designed a **hybrid ingestion system** combining **PySpark batch processing** and real-time **Kafka streaming via Docker**.
- Modeled a **Kimball methodology** with facts and dimensions for Claims analytics, using **MD5 surrogate keys** to maintain integrity.
- Implemented **HIPAA** compliance with PHI hashing and generalization, enforcing **RBAC** within **PostgreSQL** as a data warehouse.

Universal AI Assistant Chrome Extension (AI Vibecoded)

- Leveraged AI pair-programming ("**vibecoding**") to rapidly architect, iterate, and deploy a complex Chrome extension that integrates **ChatGPT**, **Claude**, and **Gemini** natively into the browser.
- Implemented Chrome's **sidePanel API** to create a seamless, persistent chat experience mirroring "Gemini in Chrome", featuring automatic session-cookie sharing so users remain authenticated without needing API keys.
- Built a **dynamic, floating UI** component that calculates `getBoundingClientRect` to seamlessly **render action buttons next to user text selections**, prioritizing a frictionless, zero-configuration user experience.

EDUCATION

University of Cincinnati
Master of Science in Information Technology

Aug 2025 – Dec 2026
Cincinnati, Ohio, USA

Anna University
Bachelor of Engineering in Electronics and Communication Engineering

Jun 2016 – Nov 2020
TamilNadu, India